ECET 250E-Linear Circuit 1
Problemset 4-Chapter 4- Circuit Theorems

Name: _____ Marks: _____

Section 4.3Superposition

1. **Chapter 4, Problem 11.** Use the superposition principle to find io and vo in the circuit of **Fig. 4.79**.

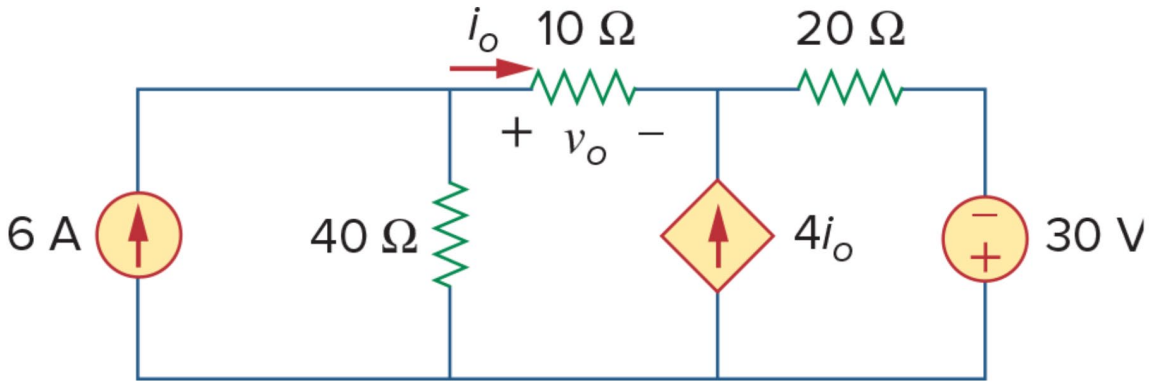


Figure 4.79 For **Prob. 4.11**.

2. **Chapter 4, Problem 12.** Determine Vo in the circuit of **Fig. 4.80** using the superposition principle.

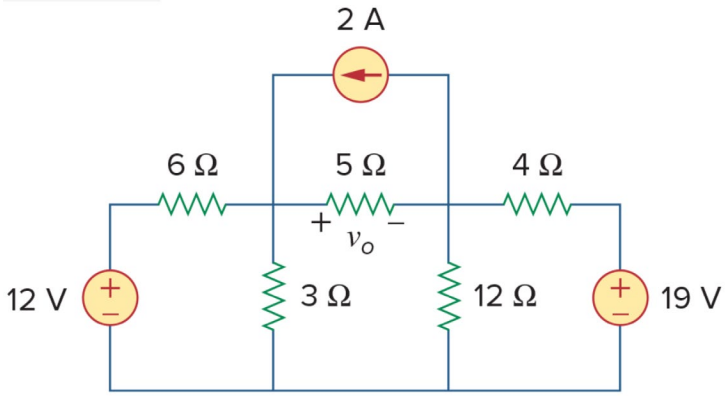


Figure 4.80 For **Prob. 4.12**.

3. **Chapter 4, Problem 16.** Given the circuit in **Fig. 4.84**, use superposition to obtain io.

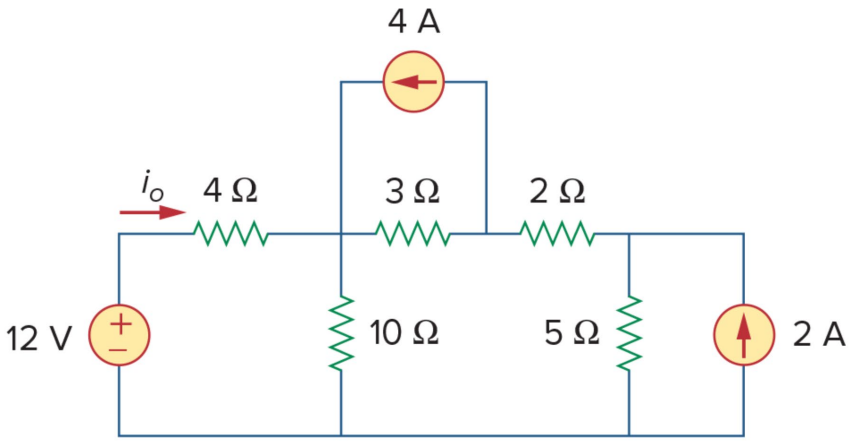


Figure 4.84 For **Prob. 4.16**.

4. Chapter 4, Problem 33. Determine the Thevenin equivalent circuit, shown in **Fig. 4.101**, as seen by the 5-ohm resistor. Then calculate the current flowing through the 5-ohm resistor.

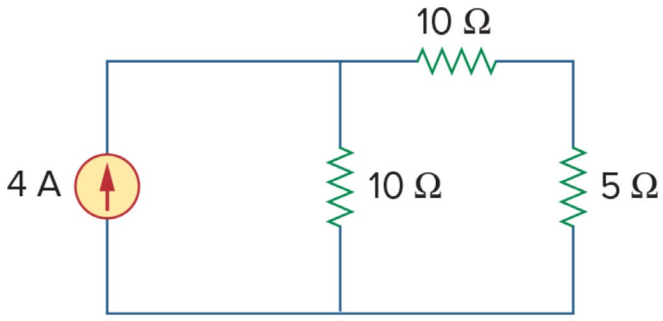


Figure 4.101 For **Prob. 4.33**.

5. **Chapter 4, Problem 37.** Find the Norton equivalent with respect to terminals *a-b* in the circuit shown in **Fig. 4.104**.

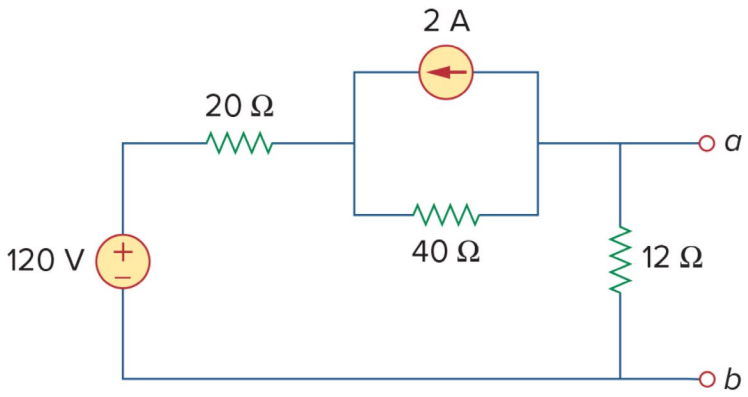


Figure 4.104 For **Prob. 4.37**.

6. **Chapter 4, Problem 36.** Solve for the current i in the circuit of **Fig. 4.103** using Thevenin's theorem. (*Hint:* Find the Thevenin equivalent seen by the 40Ω resistor.)

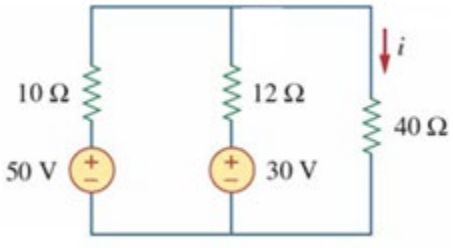


Figure 4.103 For **Prob. 4.36**.

7. **Chapter 4, Problem 38.** Apply Thevenin's theorem to find Vo in the circuit of **Fig. 4.105**.

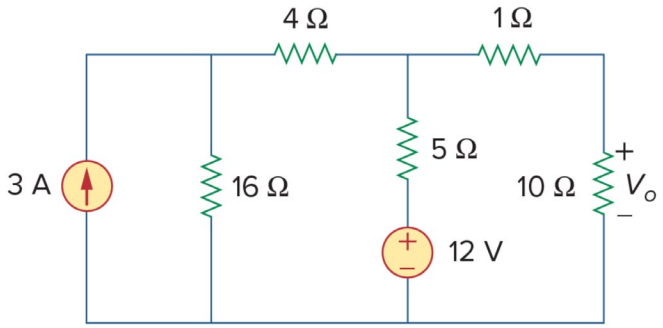


Figure 4.105 For **Prob. 4.38**.

8. **Chapter 4, Problem 39.** Obtain the Thevenin equivalent at terminals *a-b* of the circuit shown in **Fig. 4.106**.

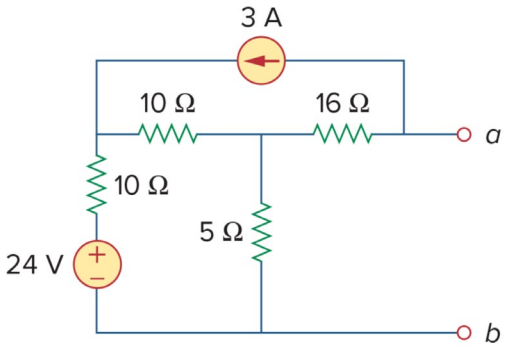


Figure 4.106 For **Prob. 4.39**.

9. **Chapter 4, Problem 41.** Find the Thevenin and Norton equivalents at terminals *a-b* of the circuit shown in **Fig. 4.108**.

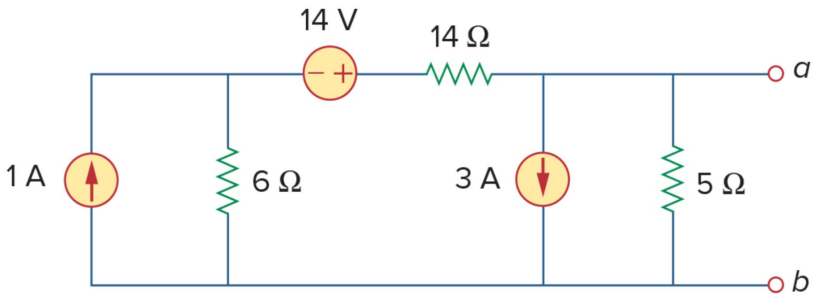


Figure 4.108 For **Prob. 4.41**.

Section 4.8Maximum Power Transfer

10. Chapter 4, Problem 66. Find the maximum power that can be delivered to the resistor *R* in the circuit of **Fig. 4.132**.

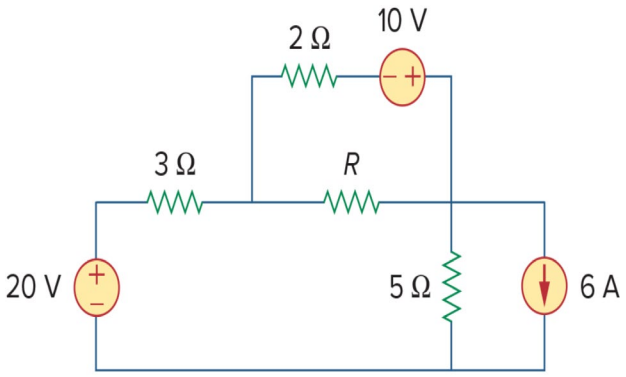


Figure 4.132 For **Prob. 4.66**.